

Margaret Ma

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EDUCATION

University of Waterloo

Candidate for Bachelor of Applied Science, Honours Electrical Engineering

- Relevant Courses: Electronic Circuits 1, Digital Computers, Signals & Systems

Waterloo, ON

Sep. 2023 – Present

PROJECTS

Quadruped Robot Platform | *Embedded Systems, ROS 2, Hardware Integration*

May 2026 – Present

- Integrated eight **LX-16A serial bus servo motors**, a **Raspberry Pi 4**, and power distribution electronics to develop a quadruped robotic platform for autonomous locomotion
- Designed robot electronics architecture including **servo communication interfaces**, **buck converter** power regulation, and battery-powered subsystems
- Deployed **ROS 2** on Raspberry Pi 4 for inter-node communication, actuator control, and hardware-software integration across robotic subsystems
- Debugged embedded controllers, **communication buses**, and robotic hardware components to improve system reliability

Autonomous Quadcopter | *ArduPilot, Pixhawk, BLDC Motors, Schematic*

Nov 2025 – Present

- Collaborated with a team to design the full system wiring schematic integrating a **Pixhawk** flight controller, **4x BLDC motors**, ESCs, GPS, RC receiver, and radio telemetry modules
- Configured power distribution architecture including **LiPo battery**, power module, and EC5 to XT60 connectors to supply flight controller and peripheral subsystems
- Integrating **ArduPilot** firmware with a **Raspberry Pi 4** companion computer for autonomous flight control and ground station communication via **Mission Planner**

STM32 Minimum System PCB | *Circuit Design, PCB Layout, STM32CubeIDE*

July 2025 – August 2025

- Designed a minimal **STM32** system PCB including **power regulation**, decoupling capacitors, and clock/reset circuitry for stable MCU initialization
- Validated **USART**, **I²C**, and **USB** peripheral interfaces in **STM32CubeIDE** through bring-up testing and debugging

EXPERIENCE

Hardware Engineering Intern, WATbotics

May 2026 – Aug 2026

University of Waterloo

Waterloo, ON

- **Designed PCBs from scratch** for autonomous robotics systems, covering schematic capture, component selection, and layout
- Researched and selected **servo motors**, evaluating torque, **communication protocols**, and compatibility with **embedded control systems**
- Configured **Raspberry Pi 4** environments and authored **setup documentation** to onboard team members onto robotics platforms
- Sourced **electronic components** and managed **procurement** to keep hardware builds on schedule

Electrical Quality Technician, Gordie Howe International Bridge

Sep 2024 – Dec 2024

Fluor Corporation

Windsor, ON

- Conducted inspections of **UPS systems**, **transformers**, **grounding networks**, and **elevator electrical rooms** on large-scale infrastructure, ensuring compliance with engineering ITPs and test plans
- Interpreted **electrical specifications and quality requirements** to verify compliance documentation against engineering standards

TECHNICAL SKILLS

Programming Languages: C++, Python, RISC-V Assembly, Matlab

Embedded Systems & Robotics: STM32CubeIDE, ROS 2, Pixhawk, MAVLink, CAN Bus, I2C, SPI, UART, Git, Linux

PCB Design & Hardware Development: Altium Designer, KiCad, LTspice, PCB Layout, Circuit Debugging

Prototyping & Test Equipment: 3D Printing, Oscilloscope, Digital Multimeter, Function Generators, Power Supply